

Name: _____
 Class: _____
 Date: _____

Mr. Rawson • GCSE Physics

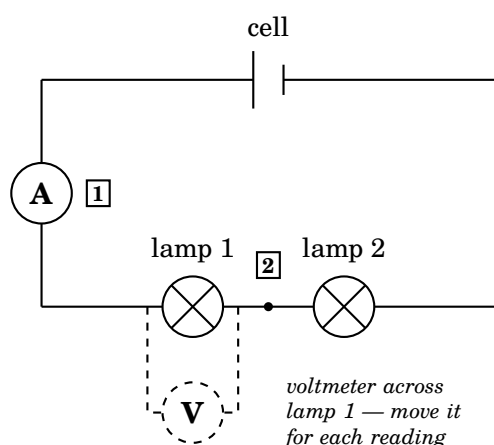
Building Series and Parallel Circuits

Key concept

There are only two ways of joining components: **series** — one single loop, so the charge has no choice of route — and **parallel** — the circuit branches at a junction, so the charge has more than one route. Whatever you measure today, $V = IR$ still applies to every single component.

Meters: the **ammeter** goes **in the loop** (in series); the **voltmeter** goes **across a component** (in parallel with it). Get these the wrong way round and the circuit will not work.

Circuit 1 — series



Method

1. Build the circuit shown. Ask before you connect the supply.
2. Record the current at position 1.
3. Move the ammeter to position 2 (between the lamps) and record again.
4. Connect the voltmeter *across* lamp 1, then lamp 2, then the cell, recording each potential difference.
5. Unscrew one lamp. Watch the other.

Position	Current / A
1	
2	

Voltmeter across	pd / V
lamp 1	
lamp 2	
the cell	

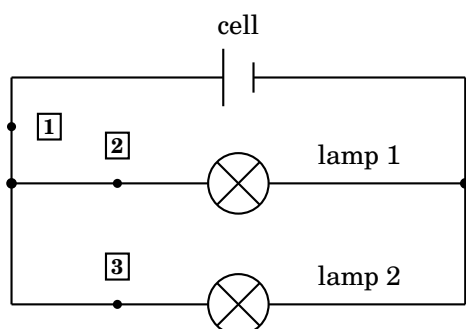
- 1) **State** what you notice about the current at different points in a series circuit. [1]

The current is the same at every point in the loop. (aP2-19 — it is not used up.)

- 2) **Compare** the pd across the two lamps with the pd across the cell. [2]

The lamp pds add up to (approximately) the cell pd — the supply's potential difference is shared between the components.

Circuit 2 — parallel



Method

1. Rebuild to this diagram: the two lamps are now on separate *branches*.
2. Put the ammeter in at position **1** (before the junction) and record. Repeat at positions **2** and **3**, one at a time.
3. Connect the voltmeter across lamp 1, then lamp 2, then the cell.
4. Unscrew one lamp. Watch the other.

Position	Current / A	Voltmeter across	pd / V
1		lamp 1	
2		lamp 2	
3		the cell	

3) State how the currents in the two branches (positions 2 and 3) relate to the current from the cell (position 1). [1]

The branch currents add up to the supply current: $I_1 = I_2 + I_3$ (aP2-20).

4) State what you notice about the pd across each lamp in the parallel circuit. [1]

The pd across each branch is the same, and equal to the supply pd.

5) Explain what happened to the other lamp when you unscrewed one — in the series circuit *and* in the parallel circuit. [2]

Series: the other lamp goes out — the single loop is broken, so no current flows anywhere. Parallel: the other lamp stays lit — its own loop with the cell is still complete.

The rules

	Series	Parallel
Current	<i>same at every point in the loop</i>	<i>branch currents add up to the supply current</i>
Potential difference	<i>shared between the components; adds up to the supply pd</i>	<i>the same across every branch — the full supply pd</i>
Adding another resistor	<i>total resistance increases: $R_{total} = R_1 + R_2$</i>	<i>total resistance decreases — an extra path for the current (no calculation needed at GCSE)</i>

6) **Calculate** the resistance of lamp 1 in your *series* circuit, using your own measured current and potential difference. [3]

Careful: V must be the pd across *that lamp alone* — not the pd across the cell.

$R = V/I$ using the lamp-1 pd and the series current. Values vary by kit — credit the rearrangement, the correct pairing of readings, and the unit (Ω).